JOHANSSON ENGINEERING

# 02 — What Equipment Do You Have

Walk your site and list every pressurised vessel. For each one, read the nameplate and write down the data. Then classify each vessel to determine what inspections it needs.

## How to Fill This In

1. Go to each vessel and find the **nameplate** (metal plate bolted to the vessel)
2. Write the nameplate data in the tables below — one row per vessel
3. Calculate **PS x V** (pressure in bar multiplied by volume in litres)
4. Use the classification guide to determine the **risk class** (A, B, or exempt)
5. Write down the last inspection date and calculate when the next one is due

**Important:** If you cannot find a nameplate, check the manufacturer's data sheet or the EU Declaration of Conformity. A vessel without identification must be assessed by an accredited inspection body before use.

## 1. Vessel Register

Write one row per vessel. **Vessel ID** — give each vessel a short code (e.g. PV-001, AR-001). **Description** — what it is (e.g. "Air receiver", "Steam boiler"). **Location** — where on site it is (e.g. "Compressor room", "Boiler house").

| NR | VESSEL ID | DESCRIPTION | LOCATION |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |
| 9 | | | |
| 10 | | | |

## 2. Technical Data

> Read these values from the **nameplate** on each vessel. **PS** = design pressure (bar). **Volume** = internal volume (litres). **PS x V** = multiply them together. **Medium** = what fluid is inside the vessel.

| NR | PS (BAR) | TEMP. MAX (C) | VOLUME (L) | PS X V | MEDIUM |
|---|---|---|---|---|---|
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | | | | |
| 5 | | | | | |
| 6 | | | | | |
| 7 | | | | | |
| 8 | | | | | |
| 9 | | | | | |
| 10 | | | | | |

**Dangerous fluids:** LPG, propane, butane, ammonia, hydrogen, chlorine, acetylene, ethylene oxide, fluorine, and other fluids classified as explosive, flammable, toxic, or oxidising under CLP Regulation (EC) 1272/2008.

**Warning:** If the medium is a dangerous fluid, the classification thresholds are much lower. Check the classification table carefully — a vessel with a dangerous fluid at PS x V = 500 is already Class B, while the same vessel with air would be exempt.

## 3. Classification

Use **PS x V** from the table above and the **medium** to determine the risk class.

## Non-dangerous fluids (Group 2)

> Includes: compressed air, nitrogen, noble gases, water, steam, hydraulic oil, and other non-hazardous fluids.

| PS X V | CLASS | WHAT IT MEANS |
|---|---|---|
| Over 10,000 | **A** | Accredited inspector required (DEKRA, Kiwa, RISE) |
| 1,001 — 10,000 | **B** | Own competent personnel may inspect |
| 51 — 1,000 | Below | Owner responsibility, no mandatory inspection |
| 50 or less | — | Not classified under AFS 2017:3 |

## Dangerous fluids (Group 1)

**Warning:** These thresholds are much lower because dangerous fluids have severe consequences if released.

| PS X V | CLASS | WHAT IT MEANS |
|---|---|---|
| Over 3,000 | **A** | Accredited inspector required |
| 201 — 3,000 | **B** | Own competent personnel may inspect |
| 51 — 200 | Below | Owner responsibility |
| 50 or less | — | Not classified under AFS 2017:3 |

**Important:** Vessels for **air and nitrogen** that would be Class B are **exempt** from periodic inspection. Mark these as "-" in the class column.

### Write your classification here

| NR | VESSEL ID | MEDIUM | GROUP | PS X V | CLASS | INSPECTOR |
|---|---|---|---|---|---|---|
| 1 | | | 1 / 2 | | A / B / - | |
| 2 | | | 1 / 2 | | A / B / - | |
| 3 | | | 1 / 2 | | A / B / - | |
| 4 | | | 1 / 2 | | A / B / - | |
| 5 | | | 1 / 2 | | A / B / - | |
| 6 | | | 1 / 2 | | A / B / - | |
| 7 | | | 1 / 2 | | A / B / - | |
| 8 | | | 1 / 2 | | A / B / - | |
| 9 | | | 1 / 2 | | A / B / - | |
| 10 | | | 1 / 2 | | A / B / - | |

**Totals:** Class A: ____ | Class B: ____ | Exempt: ____ | **Total vessels:** ____

## 4. Safety Devices

> List every safety valve, bursting disc, or pressure switch protecting your vessels. These are usually mounted directly on the vessel or on the piping close to it.

| DEVICE | TYPE | PROTECTS VESSEL | SET PRESSURE (BAR) | LAST TEST | NEXT TEST |
|---|---|---|---|---|---|
| | SV / BD / PS | | | | |
| | SV / BD / PS | | | | |
| | SV / BD / PS | | | | |
| | SV / BD / PS | | | | |
| | SV / BD / PS | | | | |

**SV** = Safety valve (spring-loaded, lifts to release pressure). **BD** = Bursting disc (ruptures at set pressure, single-use). **PS** = Pressure switch (triggers alarm or shutdown).

## 5. Placement Check

For each vessel, check that its physical location is safe. Mark OK or X. Numbers correspond to vessel Nr above.

| CHECK | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Can be maintained and inspected | | | | | |
| Protected from nearby activities | | | | | |
| Personnel can shut off if needed | | | | | |
| Personnel can evacuate if needed | | | | | |
| Temperature within design range | | | | | |

**Warning:** If any check is X, the vessel placement does not meet AFS 2017:3 2 Kap. §2-3. Document the issue and plan corrective action before the program is approved.

## 6. Document Check

For each vessel, verify that the required paperwork exists. Mark OK or X.

| CHECK | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Nameplate readable on vessel | | | | | |
| EU Declaration of Conformity on file | | | | | |
| Classification confirmed (section 3) | | | | | |
| Safety devices documented (section 4) | | | | | |
| Current inspection certificate on file | | | | | |
| Operating instructions available | | | | | |

## Prepared By

| | |
|---|---|
| **Name** | |
| **Signature** | |
| **Date** | |